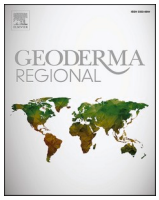
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Early-stage soil organic carbon stabilization in conservation agriculture-based cereal systems

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ABSTRACT

In South Asia, the sustainability of conventional tillage and input-intensive cereal-based cropping systems (such as rice-rice and rice-wheat) is under scrutiny due to soil organic carbon depletion, stagnant productivity, and adverse environmental impacts stemming from greenhouse gas emissions (N_2O, CH_4, and CO_2). Against this backdrop, a long-term field experiment was initiated in 2009 at three sites in India (Karnal, Patna, and Aduthurai) and one site in Bangladesh (Gazipur) to assess four scenarios (S) reflecting current and future diversified crop rotations and conservation agriculture (CA) practices: S1 - double cereal rotation with conventional practices; S2 - double cereal plus legume rotation with partial CA; S3 - double cereal plus legume rotation with full CA; and S4 - futuristic diversified cereal-legume rotations with full CA. This study delved into the dynamics and stabilization of soil organic carbon across all scenarios and sites. Replicated soil samples were collected from depths of 0–15 cm and 15–30 cm after two crop cycles. We analyzed Walkley-Black carbon (WBC), total organic carbon (TOC), and various carbon pools with different oxidizability, determining the amount of carbon stabilized under CA-based scenarios and identifying optimal systems for carbon economy in South Asia. On average, active and passive carbon pools, TOC, and WBC stocks followed the order: S4 > S3 > S2 > S1 at all sites, except Gazipur, where the order was S3 > S4 > S2 > S1. CA practices stabilized applied carbon inputs to soil organic carbon at an annual rate of 1.7 %, with greater stabilization observed under S4 (11.7 %) > S3 (10.6 %) > S2 (7.8 %), regardless of location. Rice-rice sites exhibited a higher carbon stabilization rate (13.4 %) compared to rice-wheat sites (6.7 %). A significant proportion of stabilized carbon (63 %) was allocated to passive pools in soils under S2, S3, and S4, highlighting CA's potential to enhance carbon stability. Using a linear indexing technique, we identified that both S3 and S4 are conducive to better carbon stabilization, yield sustainability, and environmental co-benefits. Consequently, full CA systems with best management practices (S3) and best management practices with crop diversification (S4) are recommended for sustainable crop production in the major double cereal growing regions of South Asia, including India and Bangladesh.

Abbreviations: IGP, Indo-Gangetic Plain; CA, Conservation agriculture; BMP, Best management practices; S, Scenarios; ZT, Zero tillage; CT, Conventional tillage; SOC, Soil organic carbon; WBC, Walkley–Black oxidizable carbon; TOC, Total organic carbon; C_{VL}, Very labile carbon; C_L, Labile carbon; C_{LL}, Less labile carbon; C_{NL}, Non labile carbon; C_{AP}, Active carbon pool; C_{PP}, Passive carbon pool; CPI, Carbon pool index; LI, Lability Index; CMI, Carbon management index; RI, Recalcitrance index; EI, Energy input; GWPi, Global warming potential intensity; REY, Rice equivalent yield.

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1. Introduction

In South Asia, rice (*Oryza sativa*), wheat (*Triticum aestivum*) and maize (*Zea mays*) are the dominant cereal crops providing the foundation of food security. The crops are commonly grown either in rotation with other non-cereal crops or in monoculture due to suitable tropical to subtropical climate of the region (Cassman, 1999; Balasubramanian et al., 2012; Erenstein et al., 2012; Jat et al., 2020; Grote et al., 2021). The rice-rice cropping system is mainly followed in Southern and Eastern parts of India, Bangladesh and Nepal (Timsina et al., 2010; Waha et al., 2020); whereas rice-wheat cropping system is extensively grown in the sub-tropical areas of the Indo-Gangetic Plains (IGP) of Bangladesh, India, Nepal, and Pakistan (Timsina and Connor, 2001; Jat et al., 2023) covering a total area of ~24Mha. In both the cases, rice is predominantly cultivated by farmers under transplanted puddled condition for better crop stand and weed management; and after harvesting of rice (for rice-wheat system), wheat is sown by conventional tillage (CT) practices. These conventional rice-rice and rice-wheat cultivation practices led to depletion of soil carbon storage, reduction of ground water table due to continuous use of flood irrigation (Mukherjee et al., 2018), occurrence of soil macro and micro nutrient deficiencies like nitrogen, phosphorus, potassium, zinc, iron and boron and overall reduction in partial factor productivity (Davis et al., 2019). Further, farmers from IGP burn rice residues for timely sowing of wheat which again deteriorates soil health and increases emission of greenhouse gases (GHGs) like carbon di oxide (Deshpande et al., 2023). Thus, there was a need for alternate crop production system which can sustain current food production capacity while maintaining or improving current soil quality. Along with yield sustainability, the crop production system may offer lower input requirement (nutrient and water), can accommodate short duration leguminous crop in between two nutrient exhaustive cereal crops and ability to recycle crop residues in order to minimize residue burning. Conservation agriculture (CA) involving the principles of minimum soil disturbance (zero tillage / reduced tillage), retention of crop residues and crop rotation with legumes had shown its potential to address all these negative effects associated with the conventional agriculture, and demonstrated a positive effect on soil carbon accrual specifically on surface soil layers (< 10 cm) (Angers and Eriksen-Hamel, 2008).

The major sources of carbon input under CA-based systems are crop residues retention, rhizodeposition carbon and inadvertent residue addition out of inclusion of crops with higher left-over residues in rotation under undisturbed soil condition (Thierfelder and Wall, 2010). A large number of researchers reported an increment, of varying magnitude, in carbon content in soils while shifting crop cultivation from conventional tillage (CT) to no tillage (NT) or reduced tillage depending upon soil and climatic conditions (West and Post, 2002; Franzluebbers, 2005; West and Six, 2007). West and Post (2002) compared data obtained from 276 paired experiment worldwide and estimated transition from CT to ZT promote carbon storage by on an average 0.57 Mg C ha^{-1} $year^{-1}$ in 0–30 cm soil depth. Under tropical humid to subhumid climatic condition, Calegari et al. (2008) reported adoption of ZT increased carbon storage by 0.24 Mg ha^{-1} $year^{-1}$ over CT after 19 years of experimentation in 0–20 cm soil depth in Brazil. In South Asian countries like India, such studies are more concentrated on rice-wheat systems in IGPs with different degree of success and carbon benefits (Grace et al., 2012; Powlson et al., 2016). For example, Sapkota et al. (2017) after seven years of experimentation reported that continuous cultivation of CT based rice-wheat system (CTR-CTW) leads to depletion of SOC (0–60 cm) by 0.9 Mg ha^{-1}; in contrast shifting from CT to ZT rice-wheat (ZTR-ZTW) sequesters 1.09 Mg C ha^{-1} (0–60 cm); again, this magnitude further increased to 4.7 Mg C ha^{-1} (0–60 cm) when residue retained along with ZT (ZTR- ZTW + R) from IGP's of eastern India. In fact, similar beneficial effect of CA under rice-wheat growing belt of IGP is well documented (Gathala et al., 2013; Jat et al., 2019; Das et al., 2013); but such studies on the effect of CA under rice-rice cropping system are scarce (Alam et al., 2015 and Sharma et al., 2019).

Long-term experiments are a useful tool for determining if a farming system can be sustained over an extended period of time in an agro-ecological zone particularly during the initial years how a plant-soil system behaves upon imposing various soil and crop management practices provides an important indication on their response toward sustainability of the system. In this background, a long -term experiment on CA with four different treatments was established at four different locations across India and Bangladesh under Cereal Systems Initiative in South Asia (CSISA) project in 2009. In these experiments, effect of CA with different drivers of agricultural changes were evaluated using existing dominant cropping systems i.e., rice-rice and rice-wheat along with a diversified cropping system on various aspects. After two years of the experimentation, Ladha et al. (2016) reported that the diversified cropping system (diversifying existing rice-rice and rice-wheat cropping system with introducing maize in rainy season, potato in winter season replacing wheat and inclusion of mungbean in summer season) had an edge over the conventional and other two CA-based systems in terms of a number of performance-indicators viz., grain yield, energy expenses, global warming potential intensity and net income. Question arises as to the robustness of these practices - are the diversified cropping systems sustainable in terms of carbon accrual and storage? Does the CA practice i.e. zero or reduced tillage + residue retention induce soil carbon build up from the beginning of the experiment under tropical to subtropical climatic conditions? If CA fails to enhance soil carbon, can it improve quality of the carbon accrued? Data regarding medium and long-term effect on CA on changes in SOC as influenced by various cropping systems, tillage practices, crop management approaches, residue utilization, fertilizer and water consumption are available from around the globe (Blanco-Canqui et al., 2021). But effect of CA on allocation of the increased C in pools of longer residence time particularly at the initial years (2–3 years) of the adoption is unknown to judge the robustness of the practices particularly under tropical to subtropical climatic condition. We hypothesize that diversified cropping scenario with higher carbon input will stabilize more carbon in soil. Keeping all these in view, an attempt was made i) to study the variation in different soil organic carbon (SOC) pools under four scenarios at four locations, ii) to study the conversion of added carbon to SOC and carbon stabilization efficiency of different CA-based cropping system, and iii) to find out the best CA options for sustainable food production and mitigation of environmental footprint.

2. Materials and methods

2.1. Experimental sites

The multi-locational production grade trials were initiated in 2009 with a long-term perspective in a regional program under the Cereal Systems Initiative for South Asia (CSISA) (http://csisa.org/). The trials were conducted at four sites in India and Bangladesh namely Indian Council of Agricultural Research (ICAR)-Central Soil Salinity Research Institute (ICAR-CSSRI), Karnal (29°70′N, 76°96E, India), ICAR-Research Complex for Eastern Region (RCER), Patna (25°24′N, 85°03′E, India), Tamil Nadu Agricultural University (TNAU), Aduthurai (11°01 N, 79°48E, India), and Bangladesh Rice Research Institute (BRRI), Gazipur (23°59′N, 90°24′E, Bangladesh) mainly to find out a sustainable cereal-based cropping system in India and Bangladesh. Improving organic carbon content in soil is fundamental for having such a sustainable system in the investigated semi-arid to sub-tropical regions. Initial soil properties and climatic variables of the study sites are given in Supplementary Table 1. Before starting of the experiment, rice crop was cultivated in all locations to obtain site uniformity.

2.2. Experimental details and management practices

Four cropping system treatments included in the study are referred to as scenarios (S). The scenarios were designed based on different drivers of agricultural changes such as rising cost of cultivation, shortages of labour and water, socioeconomic and environmental protection including carbon farming. The Scenario1 (S1) is the farmer's practice, adopted based on village surveys around each of the four sites, where CT was practiced with resources available (tillage equipments, seed, fertilizer, irrigation facility) to the farmers. In Scenario2 (S2), the crop rotation used by the farmers as in S1 was practiced with best available resources and partial CA. Scenario3 (S3) was designed with crop rotation in S1 with full CA to address issues related to degrading soil, shortages of water and labour resources. Scenario4 (S4) was designed as a diversified cropping system where main focus was on diversification of cropping pattern with CA. The details of each scenario varying from each other in crop rotation, tillage, residue management and crop establishment are presented in Table 1.

2.3. Soil sampling and analysis

Soil samples at two layers i.e., 0–15 and 15–30 cm were collected from all the three replications of the four treatments (scenarios) of the four different sites situated in different Agro-ecological zones (AEZs) of India and Bangladesh after 7 days of harvesting of *Winter* crop on completion of two cropping cycles. The samples were air dried, ground and sieved through a 2.0 mm sieve and stored. In each plot, 7 representative samples were taken randomly in a zigzag manner and pooled together to make a composite sample. Oxidizable organic carbon content of the soils (WBC) was estimated following Walkley and Black's method (Walkley and Black, 1934), while the total soil carbon (TC) was determined by CHNS analyzer (Elementer vario El Cube, Germany). As the inorganic carbon concentration was negligible, we considered total carbon as total organic carbon (TOC).

2.3.1. Analysis of pools of soil organic carbon

Modified Walkley and Black method was employed for estimating the pools of SOC using 5, 10 and 20 mL of concentrated H_2SO_4 (36 *N*) (which corresponded, respectively to 12 *N*, 18 *N* and 24 *N* of H_2SO_4) following the method of Chan et al. (2001) and separated it into very labile (C_{VL}), labile (C_L), less labile (C_{LL}) carbon pools with decreasing oxidisability. Further, the difference between TOC and carbon oxidized by 24 *N* H_2SO_4 called non-labile carbon (C_{NL} = TOC – 24 *N* H_2SO_4). The above pools are again grouped into active (C_{AP}) [$\sum(C_{VL}+ C_L)$] and passive (C_{PP}) [$\sum(C_{LL}+ C_{NL})$] pools (Mandal et al., 2007).

A lability index (LI) was also computed using the above four pools following the Eq. (1) given below (Majumder et al., 2007):

$$Lability\ Index = \frac{C_{VL}}{C_{TOC}} \times 3 + \frac{C_L}{C_{TOC}} \times 2 + \frac{C_{LL}}{C_{TOC}} \times 1 \quad (1)$$

[C_{VL}: very labile C; C_L: labile C; C_{LL}: less labile C; C_{TOC}: total organic C].

Carbon management index (CMI) was calculated using the formula as given by Blair et al. (1995).

$$CMI = \text{Carbon pool index (CPI)} \times \text{Lability index (LI)} \times 100 \quad (2)$$

Where, Carbon pool index = TOC present in field sample divided by TOC present in reference sample (Blair et al., 1995). For our experiments, we have considered farmer's practice (Sc1) as reference.

Recalcitrance index (RI) was calculated using the following Eq. (3) (Datta et al., 2017)

$$Recalcitrance\ index\ (RI) = \frac{Passive\ carbon\ pool}{Active\ carbon\ pool} \quad (3)$$

The SOC stock was calculated by the following formula.

$$\text{Carbon stock} = \text{SOC content (g/kg)} \times \text{bulk density } \left(\text{Mg m}^{-3}\right) \times \text{depth (m)} \times 10 \quad (4)$$

2.3.2. Rice equivalent yield (REY), energy input (EI) and global warming potential (GWPi)

Data of REY from different locations i.e., Karnal, Patna, Aduthurai and Gazipur were obtained from Gathala et al. (2013); Laik et al. (2014); Sharma et al. (2019) and Alam et al. (2015), respectively. Energy input (EI) and GWPi) of different scenarios located at four locations were taken from Ladha et al. (2016).

2.3.3. Carbon input calculations

The cumulative carbon input for different scenarios of conventional and CA-based cropping systems of all four locations were calculated using the crop yield and residue retention data for the experimental periods. The amount of crop residue retained in CA-based scenarios in Karnal, Patna, Aduthurai (India) and Gazipur (Bangladesh) were adapted from Gathala et al. (2013), Laik et al. (2014), Sharma et al. (2019) and Alam et al. (2015). For the calculation of plant-derived carbon i.e., rhizodeposition, root biomass carbon, carbon from stubbles or straw, we used empirical equations already published and widely accepted (Bronson et al., 1997; Mandal et al., 2007). The total amount of stubble biomass, rhizodeposition and root biomass were considered to be 2.5, 15, 15 % and 3, 12.6, 10 % of the total above-ground biomass (grain + straw) for rice and wheat at harvest, respectively (Bronson et al., 1997 and Majumder et al., 2007). The average carbon concentrations for stubble, root and rhizodeposition were 38.1, 41.2 and 74 % for rice and 35.2, 39.1 and 74 % for wheat, respectively (Majumder et al., 2007). For maize, root biomass was considered to be 29 % of total above-ground biomass and root biomass carbon was 44.4 % of total root biomass (Amos and Walters, 2006; Zhang et al., 2010). Rhizodeposition carbon from maize was calculated using a formula of CE = CR × 0.618, where CE was the estimated rhizodeposition carbon and CR was root-derived carbon (Whipps, 1985). The C input from root biomass of mungbean was assumed to be 16 % of above-ground biomass carbon (Matsumoto et al., 2021), while its rhizodeposition carbon was considered 10 % of the total root biomass (Kuzyakov and Schneckenberger, 2004). For estimating carbon inputs from potato haulm, the dry haulm yield was calculated using a harvest index of 0.7 and multiplied by its carbon concentration (36 %) (Yadav et al., 2017).

2.3.4. Carbon budgeting

The carbon build -up, total amount of carbon stabilized and allocation of stabilized carbon in different pools were calculated for 0–30 cm soil depth by the formula as given by Mandal et al. (2008); Majumder et al. (2007); Samal et al. (2017).

$$\%C\ \text{build up}\ (0-30\ \text{cm}) = \frac{(TOC_{S4}/TOC_{S3}/TOC_{S2}) - TOC_{S1}}{TOC_{S1}} \times 100 \quad (5)$$

Where, TOC_{S4}, TOC_{S3}, TOC_{S2} and TOC_{S1} indicate TOC stock in S4, S3, S2 and S1 scenarios at 0–30 cm soil depth, respectively.

Carbon stabilization % calculated using the following formula

Table 1
Scenario-wise details of cropping systems and management practices at each platform.

	Scenario 1 (Business-as-usual)	Scenario 2 (Integrated crop and resource management)	Scenario 3 (Conservation agriculture-based systems)	Scenario 4 (Diversified systems based on principles of conservation agriculture)
Drivers of change	None	Best management practice (BMP) with partial conservation agriculture (CA)	BMPs with full CA (minimum tillage+ residue retention)	BMP with crop diversification and full CA
Tillage	Conventional tillage (CT)	Combination of conventional + zero tillage (CT-ZT)	Zero, reduced and strip tillage (ZT-RT-ST)	Zero and reduced tillage (ZT-RT)
Crop rotation				
Karnal, (India)	CTR-CTW (conventional puddled transplanted rice in rainy season – conventional tillage broadcasted wheat in winter season)	CTR-ZTW-ZTMb (conventional puddled transplanted rice in rainy season – zero tillage wheat in winter – zero tillage mungbean in summer)	ZTR-ZTW-ZTMb (zero tillage rice in rainy season - zero tillage wheat in winter - zero tillage mungbean in summer)	ZTM-ZTW-ZTMb (zero tillage maize in rainy season - zero tillage wheat in winter – zero tillage mungbean in summer)
Patna, (India)	CTR-CTW	CTR-ZTW-ZTMb	ZTR-ZTW-ZTCp (zero tillage rice- zero tillage wheat - zero tillage cowpea)	ZTR - RTP - ZTM + ZTCp (zero tillage rice - reduced tillage potato - zero tillage maize grown in relay with potato and zero tillage cowpea in summer)
Aduthurai, (India)	CTR-CTR (conventional puddled transplanted rice in rainy season followed by conventional puddled transplanted rice in winter season)	CTR-RTR-ZTBg (conventional puddle transplanted rice – reduced tillage rice– zero tillage blackgram in summer)	ZTR-ZTR-ZTBg (zero tillage rice in rainy season - zero tillage rice in winter - zero tillage blackgram)	ZTM-ZTR-ZTM (zero tillage maize – zero tillage rice in winter – zero tillage maize in summer)
Gazipur, (Bangladesh)	CTR-CTR	CTR-RTR (conventional puddle transplanted rice in rainy season – reduced tillage rice in winter season)	RTR-RTR (reduced tillage rice in rainy season – reduced tillage rice in winter season)	ZTR-RTP + ZTM (zero tillage rice in rainy season – reduced tillage potato in winter – zero tillage maize in summer season)
Residue retention	No residue retention (only stubble biomass for rice and wheat left after manual harvesting)	Partial residue retention (rice: 10–20 cm anchored residues retained, wheat: 10–15 cm anchored residues retained, for legumes full residues retained)	Full residue retention (rice: 10–30 cm anchored standing residues retained, Wheat: ~ 30 cm anchored standing residues retained, For legumes full residues retained)	Full residue retention (rice: 10–30 cm anchored standing residues retained, Maize: ~ 65 cm anchored standing residues retained, Wheat: ~ 30 cm anchored standing residues retained, For legumes and potato: full residues retained)
Cultivar	Karnal: Rice- Pusa-44; Wheat- PBW 343 Patna: Rice- MTU7029 in 2009 and Arize-6444 in 2010; Wheat- PBW343 Aduthurai: Rice (rainy)- ADT 38; Rice (winter)-ADT 43, Gazipur: Rice (rainy)- BR 11 and BRRI dhan 49; Rice (winter) – BRRI dhan 28	Karnal: Rice- Arize-6444; Wheat- DBW 17; Mungbean- SML668 Patna: Rice- MTU 7029 in 2009 and Arize-6444 in 2010; Wheat- HD2733 in 2009 and PBW502 in 2010; Mungbean- SML668 Aduthurai: Rice (rainy)- ADT 46; Rice (winter)- ADT 43; Blackgram- ADT 3 Gazipur: Rice (rainy)- BRRI dhan49; Rice (winter)-BRRI dhan 28	Karnal: Rice- Arize-6129, Wheat- DBW 17 Mungbean- SML668 Patna: Rice- MTU 7029 in 2009 and Arize-6444 in 2010; Wheat- HD2733 in 2009 and PBW502 in 2010; Cowpea- Mitali Aduthurai: Rice (rainy)- ADT 46; Rice (winter)- ADT 43; Blackgram – ADT3 Gazipur: Rice (rainy)- BRRI dhan49; Rice (winter)- BRRI dhan 28	Karnal: Maize - NK-6240; Wheat- DBW 17; Mungbean: SML668 Patna: Rice- MTU 7029 in 2009 and Arize-6444 in 2010; Potato- Kufri Ashoka; Maize- Allrounder; Cowpea- Mitali Aduthurai: Rice (rainy)- ADT 38; Maize (winter)- NK 6240; Maize (summer)- NK 6240 Gazipur: Rice (rainy)- BINA 7 in 2010 and BRRI dhan 56 in 2010 Potato (winter) – Diamond, Maize- NK40, Mungbean (Summer) – BRRI mung 6
Sowing method and spacing	Broadcasting (wheat) and random transplanting (rice)	Line sowing for all crops Spacing 20 × 20 cm for rice, 22.5 × 22.5 cm for wheat, 20–30 × 20–30 cm for legumes	Line sowing for all crops Spacing 20 × 20 cm for rice, 22.5 × 22.5 cm for wheat, 20–30 × 20–30 cm for legumes	Line sowing for all crops Spacing 20 × 20 cm for rice, 22.5 × 22.5 cm for wheat, 20–30 × 20–30 cm for legumes 65 × 25 cm for potato, 60 × 20 cm for maize
Fertilizer applied	Karnal- rice and wheat: 150–58-0 kg NPK and 5 kg Zn /ha, Patna- rice: 150–11-0 NPK / ha, wheat: 150–18-17 NPK / ha, Aduthurai: rice (rainy and winter): 150–50-50 kg NPK/ha, Gazipur: rice (rainy season): 72–0-0 kg NPK / ha; winter rice: 145–18–32-12 kg NPKS /ha	Karnal- rice: 145–58-60 kg NPK and 5 kg Zn/ha, wheat: 150–58-60 kg NPK/ha, summer mungbean: residual fertilizer Patna- rice: 120–18-17 kg NPK/ha, wheat: 120–26-33 NPK / ha, summer mungbean: 15–18-0 kg NPK/ha, Aduthurai: rice (rainy and winter): 150–50-50 kg NPK/ha, summer blackgram: residual fertilizer, Gazipur: rice (rainy season): 72–16–16-8-2	Karnal - rice: 155–54-40 kg NPK and 5 kg Zn /ha; wheat: 150–58-60 kg NPK/ha, summer mungbean: residual fertilizer Patna- rice: 120–18-17 kg NPK/ha, wheat: 120–26-33 NPK / ha, cowpea: 50–26-50 kg NPK/ha, Aduthurai: rice (rainy and winter): 150–50-50 kg NPK/ha summer blakcgram: residual fertilizer Gazipur: rice (rainy season): 110–16–16-8-2 kg NPKSZn / ha; winter rice: 140–18–65-11–2 kg NPKSZn /ha	Karnal – maize: 160–60-60 kg NPK/ha; wheat: 150–58-60 kg NPK/ha, summer mungbean: residual fertilizer Patna- rice: 120–18-17 kg NPK/ha; potato: 150–40-83 kg NPK / ha; maize – 120-33-42 kg NPK/ha; cowpea: 50–26-50 kg NPK/ha, Aduthurai: rice: 150–50-50 kg NPK/ha; rainy maize-150-75-75 kg NPK/ha; winter maize-150-75-75 kg NPK/ha, Gazipur: rice (rainy season): 110–16–16-8-2 kg NPKSZn / ha, winter rice: 150–54–18.5-18-4 kg N-P-K-Zn /ha

(*continued on next page*)

Table 1 (*continued*)

	Scenario 1 (Business-as-usual)	Scenario 2 (Integrated crop and resource management)	Scenario 3 (Conservation agriculture-based systems)	Scenario 4 (Diversified systems based on principles of conservation agriculture)
		kg NPKSZn / ha; winter rice: 140–18–65-11–2 kg NPKSZn /ha		
	Fertilizers applied through urea, diammonium phosphate, muriate of potash and zinc sulfate for N, P and K and Zn respectively. 50 % N applied as basal and remaining N applied as two top dressing.			
Water management	rice: after transplanting flood irrigation 5–6 cm height maintained for 1 month, followed by water application on hairline crack	rice: after transplanting flood irrigation 5 cm height maintained for 15–20, irrigation at −40 to −50 kPa matric potential	rice: after transplanting flood irrigation 5 cm height maintained for 15–20, irrigation at −40 to −50 kPa matric potential	rice: after transplanting flood irrigation 5 cm height maintained for 15–20, irrigation at −40 to −50 kPa matric potential
	wheat: 5–6 flood irrigations applied at various growth stages like crown root initiation, tillering, flowering, milking and dough stages (Karnal, Patna and Aduthurai)	wheat: 3 irrigations applied (1 flood irrigation and 2 sprinkler irrigation; crown root initiation, tillering and flowering stages) legumes: one pre-sowing irrigation (Karnal, Patna and Aduthurai)	wheat: 3 irrigations applied (1 flood irrigation and 2 sprinkler irrigation; crown root initiation, tillering and flowering stages) legumes: one pre-sowing irrigation (Karnal, Patna and Aduthurai)	wheat: 3 irrigations applied 1 flood irrigation and 2 sprinkler irrigation; crown root initiation, tillering and flowering stages potato: 3 irrigations applied maize: Applied 2–3 irrigations on need basis,
	Gazipur: rainy season rice grown under rainfed condition. no supplemental irrigation provided	Gazipur: rainy season rice grown under rainfed condition. Need based supplemental irrigation given	Gazipur: rainy season rice grown under rainfed condition. Need based supplemental irrigation given	legumes: one pre-sowing irrigation (Karnal, Patna and Aduthurai) Gazipur: rainy season rice grown under rainfed condition. Need based supplemental irrigation given
Other management practices (weed & pest management)	Weed management- Rice: Butachlor at 1000 g ai ha^{-1} at 1 days after transplanting (DAT) fb hand weeding to remove escaped weeds (Karrnal and Patna)	Weed management- Butachlor at 1000 g ai ha^{-1} at 1 DAT fb hand weeding to remove escaped weeds (Karrnal and Patna)	Weed management- Oxadirgyl 90 g ai ha^{-1} or pendimethalin at 1000 g ai ha^{-1} as pre-emergence (PRE) at 2–3 DAS 'fb' azimsulfuron at 35 g ai ha^{-1}	Weed management- Maize: Atrazine was applied either as PRE at 625 g ai ha^{-1} at 1 DAS in first year
	Aduthurai: hand weeding at 10, 30 and 60 DAT	Aduthurai: cono weeder operation was given 10 and 20 DAT. One 60 hand weeding was done DAT	Aduthurai: If needed hand weeding according to the weed growth	Aduthurai: If needed hand weeding according to the weed growth
	Gazipur: Pendimethalin applied as PRE	Gazipur: Pendimethalin applied as early post emergence 2–3 DAT,	Gazipur: Pendimethalin applied as early post emergence 2–3 DAT,	Gazipur: Pendimethalin applied as early post emergence 2–3 DAT,
	Wheat: tank mix Clodinafop + Metsulfuronmethyl (Karnal) or Metsulfuronmethyl + sulfosulfuron (Patna) at 25–35 DAS	Wheat: tank mix Clodinafop + Metsulfuronmethyl (Karnal) or Metsulfuronmethyl + sulfosulfuron (Patna) at 25–35 DAS	Wheat: tank mix Clodinafop + Metsulfuronmethyl (Karnal) or Metsulfuronmethyl + sulfosulfuron (Patna) at 25–35 DAS	Wheat: tank mix Clodinafop + Metsulfuronmethyl 25–35 DAS (Karnal) Potato- Hand weeding
Seed treatment	Rice Karnal and Patna: tebuconazole + imidacloprid @ 1+ 2 mL / kg seed Gazipur: no seed treatment Aduthurai: Pseudomonas fuorescens @ 10 g/kg seed Wheat: tebuconazole @ 1 mL /kg seed or mancozeb 2 g /kg seed	Rice Karnal and Patna: tebuconazole + imidacloprid @ 1+ 2 mL / kg seed Gazipur: bavistine @ 3 g/kg seed Aduthurai: Pseudomonas fuorescens @ 10 g/kg seed Wheat: tebuconazole @ 1 mL /kg seed or mancozeb 2 g /kg seed	Rice Karnal and Patna: tebuconazole + imidacloprid @ 1+ 2 mL / kg seed Gazipur: bavistine @3 g /kg seed Aduthurai: Pseudomonas fuorescens @ 10 g/kg seed Wheat: tebuconazole @ 1 mL /kg seed or mancozeb 2 g /kg seed	Rice Karnal and Patna: tebuconazole + imidacloprid @ 1+ 2 mL / kg seed Gazipur: bavistine @ 3 g/kg seed Aduthurai: Pseudomonas fuorescens @ 10 g/kg seed Wheat: tebuconazole @ 1 ml /kg seed or mancozeb 2 g /kg seed Potato: Mancozeb @ 2 g /kg potato seed

$$\%\text{C stabilization } (0-30\text{ cm}) = \frac{\text{Carbon buildup over S1}}{\text{Additional C input gained over S1}} \times 100 \quad (6)$$

While the allocation of stabilized carbon in active and passive pools were calculated using the following formula,

Allocation of stabilized carbon in active or passive pool

$$\%\text{C stabilized in active or passive pools } (0-30\text{ cm}) = \frac{[(C_{S4}/C_{S3}/C_{S2})–C_{S1}] \text{ in active or passive pool}}{\text{carbon build up in S4/S3/S2 over S1}} \times 100 \quad (7)$$

Where, $C_{S4/S3/S2/S1}$ are active and passive pool carbon stock present in S4, S3, S2 and S1 scenarios at 0–30 soil depth, respectively.

2.3.5. Optimization of the management practices

We made an attempt to evaluate the relative performance of the CA-based management practices in different locations. The performance of the scenarios was assessed using linear indexing technique for the goodness of the system (Liebig et al., 2001). To this end, firstly the attributes/parameters were separated into two groups: one is the gain-parameters for which 'more is better', and it includes rice equivalent yield (REY), total organic carbon stabilized (%) and % passive carbon to TOC. The other is the loss-parameters for which 'less is better', and includes global warming potential intensity (GWPi), energy input (EI). Values measured for each of the gain-parameters for the four scenarios were divided by its highest observed value so that the scenario with the highest value received a score between zero to one (Eq. (7)). Contrarily, values measured for each of the loss-parameters of the scenario were divided by its lowest observed value and made reciprocal so that the lowest observed value received a score between zero to one (Eq. (8)). Following such scoring for each of the parameters for all the scenarios, we summed up the scores and ranked the scenarios.

$$\text{Score}_{(\text{REY,\%PP/TOC,\%C stabilization})} = \frac{Xi}{XMax} \quad (8)$$

$$\text{Score}_{(\text{EI and GWPi})} = \frac{XMin}{Xi} \quad (9)$$

Total Score = Σ (Score obtained for "more is better" variables + Score obtained from "less is better" variables).

2.3.6. Statistical analysis

Statistical analysis of the results was performed using one way Analysis of Variance (ANOVA) technique at 0.05 % level of significance. For every parameter, normality of data and homogeneity of variances between the scenarios have been tested using Shapiro-wilk test and Bartlett's test, respectively. Duncan's multiple-range test (DMRT) was used to compare treatment means. These analyses along with simple correlation between the soil parameters were computed using Windows-based SPSS program Version 17.0, SPSS, Chicago, IL.

3. Results

3.1. Carbon content in soil under different scenarios

Adoption of CA i.e., zero or reduced tillage, diversified crop rotation and residue retention had a positive effect on WBC and TOC stock at 0–15 cm soil layer, but not in 15–30 cm. At surface soil (0–15 cm), highest WBC and TOC stock were found in S4 among all the locations except Gazipur site, where highest stocks were observed under S3. When we compared WBC and TOC stock for 0–30 cm soil depth; WBC stock remain unchanged, but TOC stock showed marked variation among various scenarios (Table 2). On average, higher TOC stock was found in soils under CA based scenarios S4 (38.73 Mg ha^{-1}), S3 (38.55 Mg ha^{-1}), S2 (38.41 Mg ha^{-1}) over S1 at 0–30 cm soil depth irrespective of locations (Table 2).

Allocation of TOC in different pools of oxidizability also followed similar trend (Table 3). Across the locations in surface soil, the amount of C_{VL} and C_{NL} pools were higher with S4 (8.25 and 4.29 Mg ha^{-1}) and S3 (8.29 and 4.23 Mg ha^{-1}) than those with S2 (8.07 and 4.02 Mg ha^{-1}) and S1 (7.98 and 3.97 Mg ha^{-1}) except Gazipur (where C_{VL} is slightly lower in S3 over S2 and S1); while the C_L and C_{LL} pools did not show any definite trend. In subsurface soil (15–30 cm), the distribution of carbon pools did not follow any pattern unlike surface soil. But, when 0–30 cm soil depth was considered distribution of carbon pools followed similar trend as observed in surface soil layer. Both the active (C_{AP}) and passive pools (C_{PP}), in general, recorded an increasing trend along the scenarios, Sc4 (23.97 and 15.82 Mg ha^{-1}) > Sc3 (23.85 and 15.63 Mg ha^{-1}) > Sc2 (23.77 and 15.50 Mg ha^{-1}) > Sc1 (23.63 and 15.24 Mg ha^{-1}) in 0–30 cm soil layer (Supplementary Table 3).

3.2. Carbon build up, stabilization and allocation of stabilized carbon into various pools

On average, at 0–30 cm soil depth, TOC and passive carbon pool build-up under CA based scenarios over S1 irrespective of locations followed the order of S4 (0.92 Mg ha^{-1}) > S3 (0.61 Mg ha^{-1}) > S2 (0.41 Mg ha^{-1}); with highest (1.02 Mg ha^{-1}) observed in Karnal site. The range of TOC build up under CA based scenarios (S2, S3 and S4) in Karnal, Patna, Aduthurai and Gazipur varied from 2.38 to 3.51, 0.64–2.66, 0.87–1.52 and 0.67–2.27 %, respectively (Supplementary Table 3).

We obtained a positive correlation between TOC build up over S1 and additional carbon input received over S1 ($r = 0.69$, $p < 0.05$). A linear regression was observed; $y = 0.034\times + 0.388$; $R^2 = 0.48$, $p < 0.05$ where y = change in TOC stock over two years, x = additional carbon input received over S1 in two years. The slope (3.4 %) of the curve for the relationship represented an average conversion/stabilization rate (over two year) of the additional carbon inputs into TOC under the tested CA-based cropping systems (Fig. 1). When we calculated such stabilization of additional carbon inputs to TOC gain over S1 for different locations and scenarios, we found it (conversion rate) higher in soils of Gazipur (14.7 %) and Aduthurai (12.1 %), but lower in Karnal (7.6 %) and Patna (5.8 %) (Table 5). Among the scenarios, stabilization of applied carbon input was found in the order of S4 (11.7 %) > S3 (10.6

Table 2
Depth-wise distribution of Walkley –Black carbon (WBC) and total organic carbon (TOC) stock (Mg ha^{-1}) at different CA based scenarios located at various sites.

Research platforms	Scs	WBC stock			TOC stock		
		0–15 cm	15–30 cm	0–30 cm	0–15 cm	15–30 cm	0–30 cm
Karnal	S1	12.70 ± 0.06b	12.47 ± 0.14a	25.17 ± 0.25a	15.17 ± 0.09a	13.90 ± 0.13a	29.07 ± 0.08b
	S2	13.28 ± 0.11a	12.38 ± 0.16a	25.66 ± 0.12a	15.76 ± 0.14ab	13.99 ± 0.17a	29.75 ± 0.16ab
	S3	13.26 ± 0.05a	12.15 ± 0.24a	25.41 ± 26a	16.11 ± 0.10a	13.76 ± 0.18a	29.87 ± 0.14ab
	S4	13.32 ± 0.06a	12.14 ± 0.16a	25.46 ± 0.27a	16.24 ± 0.15a	13.84 ± 0.13a	30.08 ± 0.13a
Patna	S1	16.62 ± 0.22b	15.51 ± 0.11a	32.14 ± 0.34a	20.56 ± 0.30b	17.02 ± 0.15a	37.58 ± 0.45b
	S2	16.67 ± 0.14b	15.47 ± 0.27a	32.13 ± 0.50a	20.68 ± 0.16b	17.14 ± 0.35a	37.82 ± 0.50ab
	S3	17.05 ± 0.17a	15.25 ± 0.23a	32.30 ± 0.14a	21.17 ± 0.18a	16.95 ± 0.05a	38.12 ± 0.14ab
	S4	17.29 ± 0.06a	15.31 ± 0.18a	32.60 ± 0.25a	21.52 ± 0.11a	17.06 ± 0.17a	38.58 ± 0.24a
Aduthurai	S1	18.97 ± 0.12b	18.98 ± 0.28a	37.95 ± 0.35a	23.95 ± 0.17b	22.73 ± 0.34a	46.68 ± 0.32b
	S2	19.27 ± 0.16ab	18.99 ± 0.17a	38.26 ± 0.22a	24.34 ± 0.09ab	22.75 ± 0.20a	47.09 ± 0.18a
	S3	19.55 ± 0.10a	18.64 ± 0.12a	38.20 ± 0.24a	24.76 ± 0.07a	22.41 ± 0.21a	47.17 ± 0.28a
	S4	19.61 ± 0.08a	18.71 ± 0.06a	38.33 ± 0.18a	24.78 ± 0.22a	22.62 ± 0.18a	47.40 ± 0.24a
Gazipur	S1	18.91 ± 0.06b	13.56 ± 0.16a	32.48 ± 0.18a	23.41 ± 0.03b	18.73 ± 0.16a	42.14 ± 0.18b
	S2	19.02 ± 0.06ab	13.67 ± 0.12a	32.69 ± 0.14a	23.55 ± 0.10ab	18.87 ± 0.12a	42.42 ± 0.09ab
	S3	19.24 ± 0.05a	13.37 ± 0.18a	32.61 ± 0.09a	23.96 ± 0.21a	18.79 ± 0.09a	42.75 ± 0.34ab
	S4	19.21 ± 0.07a	13.51 ± 0.20a	32.72 ± 0.12a	24.07 ± 0.14a	19.02 ± 0.09a	43.09 ± 0.06a

Values with different lower case (a–c) letters are significantly different between each scenario at $P < 0.05$ (DMRT test performed for separation of mean); ± values indicate standard error of mean of the observed values (WBC: Walkley-Black Carbon; TOC: Total organic carbon).
Where, S1: Conventional rice-wheat or rice-rice system with conventional farmer's practice; S2: Partial CA based rice-wheat-mungbean or rice-rice-blackgram or rice-rice-fallow system with best management practice (BMPs); S3: Full CA based rice-wheat-mungbean or rice-rice-blackgram or rice-rice-fallow with BMPs; S4 – Full CA based maize-wheat-mungbean or rice-potato – maize + mungbean or maize-rice-maize or rice-potato- maize system with BMPs.

Table 3
Distribution of soil organic carbon in different pools under conservation agriculture practices at various research platforms.

Locations	Scs	Soil organic carbon pools of different oxidizabilities (Mg ha^{-1})							
		Very labile carbon (C_{VL})		Labile carbon (C_L)		Less labile carbon (C_{LL})		Non labile carbon (C_{NL})	
		0–15 cm	15–30 cm	0–15 cm	15–30 cm	0–15 cm	15–30 cm	0–15 cm	15–30 cm
Karnal	S1	5.74 ± 0.48b	5.19 ± 0.12b	4.56 ± 0.19a	5.10 ± 0.06a	2.40 ± 0.17a	2.19 ± 0.05a	2.47 ± 0.06b	1.42 ± 0.07b
	S2	6.30 ± 0.09a	5.44 ± 0.06a	4.29 ± 0.20a	4.80 ± 0.07ab	2.69 ± 0.16a	2.14 ± 0.02a	2.48 ± 0.04b	1.62 ± 0.05a
	S3	6.50 ± 0.13a	5.57 ± 0.03a	4.21 ± 0.17a	4.56 ± 0.12b	2.55 ± 0.09a	2.02 ± 0.03b	2.85 ± 0.12a	1.61 ± 0.06a
	S4	6.43 ± 0.09a	5.69 ± 0.07a	4.45 ± 0.13a	4.41 ± 0.05b	2.44 ± 0.12a	2.04 ± 0.05b	2.92 ± 0.09a	1.70 ± 0.05a
Patna	S1	9.24 ± 0.12b	5.60 ± 0.17b	4.26 ± 0.15a	5.82 ± 0.19a	3.12 ± 0.5b	4.10 ± 0.14a	3.94 ± 0.07b	1.50 ± 0.04b
	S2	9.10 ± 0.08b	6.53 ± 0.20a	4.35 ± 0.12a	5.03 ± 0.10b	3.21 ± 0.02a	3.91 ± 0.03b	4.01 ± 0.07ab	1.67 ± 0.07a
	S3	9.64 ± 0.15a	6.31 ± 0.07a	4.15 ± 0.10a	5.05 ± 0.05b	3.26 ± 0.05a	3.89 ± 0.02b	4.12 ± 0.01a	1.70 ± 0.05a
	S4	9.74 ± 0.03a	5.81 ± 0.19b	4.12 ± 0.11a	5.65 ± 0.11a	3.43 ± 0.15a	3.84 ± 0.09b	4.23 ± 0.05a	1.75 ± 0.02a
Aduthurai	S1	7.64 ± 0.05a	6.30 ± 0.16a	5.26 ± 0.06b	5.84 ± 0.05b	6.07 ± 0.15a	6.85 ± 0.08a	4.99 ± 0.01b	3.75 ± 0.06a
	S2	7.42 ± 0.05a	6.11 ± 0.07a	5.63 ± 0.05b	6.00 ± 0.03a	6.22 ± 0.06a	6.88 ± 0.06a	5.08 ± 0.04b	3.75 ± 0.06a
	S3	7.75 ± 0.06a	6.01 ± 0.05a	5.67 ± 0.07b	5.77 ± 0.04b	6.13 ± 0.11a	6.86 ± 0.10a	5.21 ± 0.05a	3.76 ± 0.08a
	S4	7.08 ± 0.07b	6.06 ± 0.07a	6.22 ± 0.07a	5.94 ± 0.09a	6.32 ± 0.18a	6.71 ± 0.08a	5.17 ± 0.05a	3.90 ± 0.09a
Gazipur	S1	9.32 ± 0.05b	3.92 ± 0.05b	4.99 ± 0.04b	4.4 ± 0.09a	4.60 ± 0.15a	3.92 ± 0.12a	4.49 ± 0.08b	5.17 ± 0.15b
	S2	9.46 ± 0.04a	5.01 ± 0.06a	4.85 ± 0.07a	4.77 ± 0.04a	4.71 ± 0.09a	3.89 ± 0.11a	4.52 ± 0.04b	5.21 ± 0.07b
	S3	9.25 ± 0.06b	4.94 ± 017a	5.15 ± 0.02a	4.86 ± 0.08a	4.84 ± 0.20a	3.57 ± 0.19a	4.72 ± 0.08a	5.42 ± 0.06a
	S4	9.75 ± 0.08a	5.72 ± 0.09a	4.89 ± 0.05b	3.90 ± 0.011b	4.57 ± 0.11a	3.89 ± 0.11a	4.86 ± 0.16a	5.51 ± 0.06a

Values with different lower case (a–c) letters are significantly different between each scenario at $P < 0.05$ (DMRT test performed for separation of mean); ± values indicate standard error of mean of the observed values.
Where, S1: Conventional rice-wheat or rice-rice system with conventional farmers practice; S2: Partial CA based rice-wheat-mungbean or rice-rice-blackgram or rice-rice-fallow system with best management practice (BMPs); S3: Full CA based rice-wheat-mungbean or rice-rice-blackgram or rice-rice-fallow with BMPs; S4 – Full CA based maize-wheat-mungbean or rice-potato – maize + mungbean or maize-rice-maize or rice-potato- maize system with BMPs.

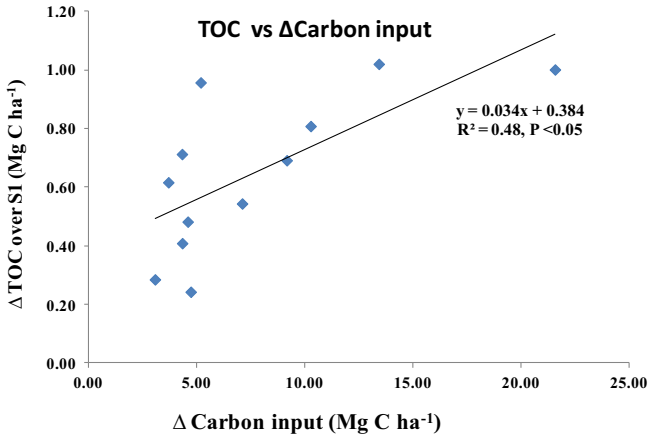


Fig. 1. Relationship between change in additional carbon inputs received vs change in TOC over S1.
(Where, S1: Conventional rice-wheat or rice-rice system with conventional farmers practice; S2: Partial CA based rice-wheat-mungbean or rice-rice-blackgram or rice-rice-fallow system with best management practice (BMPs); S3: Full CA based rice-wheat-mungbean or rice-rice-blackgram or rice-rice-fallow with BMPs; S4 – Full CA based maize-wheat-mungbean or rice-potato – maize + mungbean or maize-rice-maize or rice-potato-maize system with BMPs.
ΔTOC: Change in total organic carbon stock in CA based S2/S3/S4 as compared to S1; ΔCarbon input: additional carbon input gain in CA based S2/S3/S4 over S1).

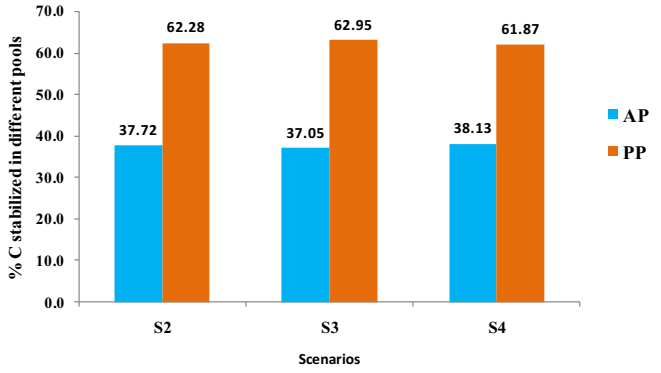


Fig. 2. Variations in the allocation of stabilized carbon in active and passive pools in soils under different scenarios.
(Where, AP: Active carbon pool; PP: Passive carbon pool).

%) > S2 (7.8 %) irrespective of locations (Table 5).

We found that applied carbon got stabilized more in passive pool with a concomitant reduction in active pools in all the CA based scenarios; % TOC stabilized in passive pool comprises more than 60 % in all the sites with slight or no variation among them (Supplementary Table 3 and Fig. 2). Similar trend was also observed in allocation of stabilized carbon in passive pools among various CA based scenarios; all the CA based scenarios (S2, S3 and S4) stabilized 60 % or more in passive carbon pool over S1 (Supplementary Table 3 and Fig. 2).

3.3. Recalcitrance index (RI), carbon management index (CMI) and lability index (LI) under various scenarios

On average, the magnitudes of RI values were higher with Sc3 and Sc4 over Sc1 concomitant with a lower value of LI particularly with Sc4 soils (Supplementary Table 4). This was more pronounced in soils of Gazipur and Aduthurai compared with Karnal and Patna. There was inconsistent trend of these values (RI and LI) in sub-surface layer (15–30 cm). The CMI was higher with Sc3 and Sc4 compared with Sc2 (in both soil depths), and this was more so in Karnal and Patna sites.

3.4. Optimizing the systems performance and identification of best management practices

We tried to optimize the systems taking into consideration their productivity (REY), global warming potential intensity (GWPi), energy expenses, carbon stabilization (total C stabilized), % passive carbon to total TOC, and carbon conversion efficiencies. On the basis of calculation of scores, on average, the relative performance of the scenarios, irrespective of the sites, followed the order: S4 (3.80) > S3 (3.60) > S2 (3.08) > S1 (1.94) (Fig. 3 and Supplementary Table 5). This was true for all the sites except Patna where S3 excelled over the S4 (Supplementary Table 5).

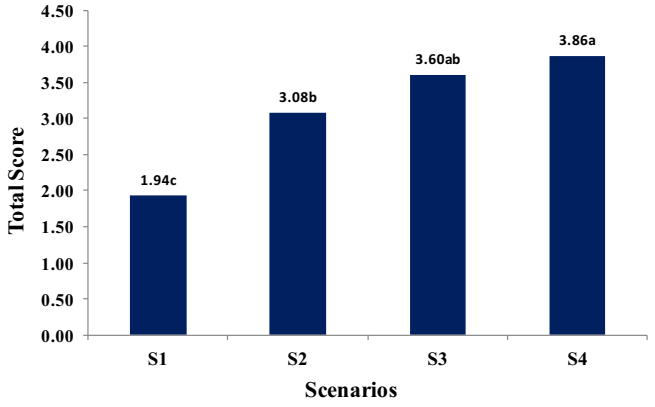


Fig. 3. Overall system score of various cropping system scenarios under conservation agriculture practices.
Values with different lower case (a–c) letters are significantly different between each scenario at $P < 0.05$ (DMRT test performed for separation of mean (Where, S1: Conventional rice-wheat or rice-rice system with conventional farmers practice; S2: Partial CA based rice-wheat-mungbean or rice-rice-blackgram or rice-rice-fallow system with best management practice (BMPs); S3: Full CA based rice-wheat-mungbean or rice-rice-blackgram or rice-rice-fallow with BMPs; S4 – Full CA based maize-wheat-mungbean or rice-potato – maize + mungbean or maize-rice-maize or rice-potato-maize system with BMPs.)

4. Discussion

4.1. Variation in carbon stock and stabilization

Our results clearly indicate that CA coupled with BMPs (S3) and BMPs + diversified scenarios (S4) showed higher TOC and WBC irrespective of cropping systems and locations of the experiment. This was because of combination of ZT and residue retention; which minimizes exposure of undisturbed SOC to oxidation and higher carbon input through crop residues. CT practice under S1 exposed soil aggregates to air, favouring higher carbon oxidation from soil; in contrast, zero tillage under S3 and S4 reduced the exposure of soil to air which ultimately helped to cut down the CO_2 emission and helped in higher TOC build-up (Abdalla et al., 2016; Wang et al., 2020; Cooper et al., 2021). Apart from this, on average higher amount of carbon inputs (through rhizodeposition, root biomass, and crop residue) were observed in CA based scenarios (S4: 17.9 Mg ha^{-1}, S3: 13.1 Mg ha^{-1}, and S2: 12.1 Mg ha^{-1}) compared to S1 (6.7 Mg ha^{-1}) scenario in all the sites (Table 4). Like others (Wang et al., 2016; Luo et al., 2017; Samal et al., 2017), we expected that this higher amount of C inputs would cause proportional increase in TOC in soils under S4, S3 and S2 compared to S1 (Table 4) scenario. Inclusion of legumes in summer season, especially mungbean and cowpea, and incorporating their residues further supports higher TOC in S4, S3 and S2 over S1 (Hazra et al., 2020). Legume residues with a relatively narrow C:N ratio decomposed quickly in soil enhancing soil microbial activity and help conversion of crop residue carbon into SOC, and further support higher carbon build-up in S3 and S4 over S1.

The SOC fractions allocated into different pools based on the gradient of oxidizability, varied with their chemical composition. We observed a higher proportion of carbon allocated to very labile carbon pool (C_{VL}) followed by less labile pool (C_{LL}) and non labile pool (C_{NL}) in S3 and S4 at surface soil layer. Retention of higher quantities of rice, wheat and maize residues with relatively wider C:N ratios slows down the carbon mineralization from crop residues (Datta et al., 2019) and gradually releases secondary substrates, which ultimately enhanced the C_{VL} in surface soil; on the other hand, undecomposed portion of the crop residues increased the formation of less labile carbon (C_{LL}) and non labile carbon (C_{NL}) pool. It was reported that organic carbon derived from crop residues was first allocated into labile carbon pools and gradually accumulated; later, it enriched with stable and recalcitrant carbon fractions in soil (Six et al., 2002). This increasing trend of C_{NL} and C_{PP}, especially at surface soil layer (0–15 cm) of S3 and S4 clearly indicates that CA practices (ZT, residue retention + diversified crop rotation) might have induced recalcitrance nature of soil carbon. The elimination of CT and subsequently, an enhancement of less labile (C_{LL}) and non-labile carbon (C_{NL}) pools under CA is also reported by Nandan et al. (2021) in a cereal-based cropping system. Similarly, Sharma et al. (2019) also recorded higher passive pool carbon under rice and wheat residue retention treatments compared to no residue retention in North West India in a rice-wheat cropping system. The highest proportion of C_{NL} and C_{PP} in S4 at the surface soil irrespective of the experimental sites may be due to the higher biochemical recalcitrance of the added maize residues over that of rice and wheat (Buranov and Mazza, 2008; Jat et al., 2019). Again, maize root with higher lignin content (Abiven et al., 2011) secrets gel-like root exudates in soil (Naveed et al., 2017) which facilitates formation of more stable carbon forms in the surrounding soil,

Table 4
Rice equivalent yield, cumulative crop residues recycled and estimated carbon input from various sources.

Research platforms	Scs	Rice equivalent yield (two-year mean data) (Mg ha^{-1})	Cumulative Crop residues recycled (Mg ha^{-1})	Cumulative carbon input (CI) (Mg ha^{-1})			
				Root carbon	Rhizodeposition carbon	Residue carbon	Total
	S1	12.3 ± 0.13d	0	2.31 ± 0.04c	4.67 ± 0.06c	0.46 ± 0.05b	7.4 ± 0.03c
	S2	15.7 ± 0.18a	17.6 ± 0.10	3.65 ± 0.04b	6.41 ± 0.07a	6.56 ± 0.10a	16.6 ± 0.12b
	S3	14.4 ± 0.12b	22.4 ± 0.14	3.16 ± 0.04b	5.69 ± 0.05b	8.88 ± 0.09a	17.7 ± 0.10b
Karnal	S4	13.4 ± 0.16c	22.1 ± 0.16	6.22 ± 0.14a	5.79 ± 0.05b	8.88 ± 0.04a	20.9 ± 0.17a
	S1	8.5 ± 0.12c	0.0	2.00 ± 0.04b	3.90 ± 0.10c	0.37 ± 0.02b	6.3 ± 0.06c
	S2	15.1 ± 0.14b	9.1 ± 0.09	2.69 ± 0.06b	4.68 ± 0.07b	3.63 ± 0.11b	11.0 ± 0.07b
	S3	13.0 ± 0.09b	12.8 ± 0.18	3.01 ± 0.09b	5.25 ± 0.11b	5.12 ± 0.10b	13.4 ± 0.12b
Patna	S4	21.5 ± 0.31a	37.5 ± 0.21	6.84 ± 0.04a	6.58 ± 0.10a	14.45 ± 0.09a	27.9 ± 0.11a
	S1	7.4 ± 0.13c	0	2.01 ± 0.04b	3.63 ± 0.07b	0.31 ± 0.03c	5.9 ± 0.08b
	S2	10.5 ± 0.13a	7.1 ± 0.08	2.83 ± 0.08a	4.69 ± 0.06a	2.78 ± 0.08b	10.3 ± 0.10a
	S3	8.6 ± 0.28b	11.4 ± 0.12	2.31 ± 0.09a	3.83 ± 0.05b	4.40 ± 0.11b	10.5 ± 0.07a
Aduthurai	S4	6.0 ± 0.14d	12.7 ± 0.16	2.53 ± 0.06a	2.84 ± 0.05c	4.91 ± 0.09b	10.3 ± 0.06a
	S1	5.0 ± 0.08c	0	2.43 ± 0.02b	4.39 ± 0.06b	0.38 ± 0.05c	7.2 ± 0.10b
	S2	6.3 ± 0.14b	5.4 ± 0.10	2.90 ± 0.09b	5.23 ± 0.06a	2.16 ± 0.10b	10.3 ± 0.12a
	S3	6.2 ± 0.13b	5.8 ± 0.09	3.06 ± 0.09b	5.53 ± 0.07a	2.32 ± 0.10b	10.9 ± 0.11a
Gazipur	S4	8.5 ± 0.08a	7.9 ± 0.11	4.13 ± 0.10a	4.15 ± 0.10b	4.12 ± 0.08a	12.4 ± 0.09a

Values with different lower case (a–c) letters are significantly different between each scenario at P < 0.05 (DMRT test performed for separation of mean); ± values indicate standard error of mean of the observed values.
Where, S1: Conventional rice-wheat or rice-rice system with conventional farmers practice; S2: Partial CA based rice-wheat-mungbean or rice-rice-blackgram or rice-rice-fallow system with best management practice (BMPs); S3: Full CA based rice-wheat-mungbean or rice-rice-blackgram or rice-rice-fallow with BMPs; S4 – Full CA based maize-wheat-mungbean or rice-potato – maize + mungbean or maize-rice-maize or rice-potato- maize system with BMPs.

thereby accentuating higher passive and recalcitrant carbon build-up in S4.

4.2. Carbon build up, stabilization and its allocation

The net gain of SOC over initial SOC content is called carbon sequestration or storage; whereas carbon build up is used to compare carbon gain between two or more treatments based on their present status (Mandal et al., 2007; Samal et al., 2017). In our study we obtained relatively greater carbon build up in CA based scenarios (S2, S3 and S4) over conventional one occurred under Karnal site followed by Patna site respectively. This quick gain in TOC among those sites was linked with lower initial SOC status (Supplementary table 1) and favourable land management practices (zero tillage and residue retention) (Arndt et al., 2022). Recently, Slessarev et al. (2023) compared multiple cross site investigations and reported that SOC-poor soils tend to absorb or add carbon more quickly than SOC-rich soils.

A number of interrelated elements, such as climate (temperature and rainfall), the amount of carbon input received, substrate biochemistry, soil properties (texture), systemic tillage intensity, and the surrounding environment, interact to control the stabilization of additional organic carbon in soil (Kong et al., 2005; Schmidt et al., 2011). In the present experiment, overall, 3.4 % additional carbon input received through crop residue retention, root biomass and rhizodeposition got stabilized into TOC (0–30 cm) over two years (Fig. 1) irrespective of research locations. However, the observed conversion efficiencies of the carbon input to TOC (~ 3.4 % over two years or 1.7 % annually) are lower than those reported by others from subtropical climatic conditions (Zhang et al., 2010; Liang et al., 2023). We may have noticed a lower conversion value because of the hot and humid temperature at our experimental sites, which aided in the faster oxidation of additional residual organic carbon.

Among the various locations, the carbon stabilization % varied from 5.8 to 14.7 % (over two years). On average, all these values of conversion of residue carbon to TOC were higher or within the range as reported from this part of the world by others (Kong et al., 2005; Jiang et al., 2018), possibly because of a higher carbon saturation deficit in the soils of the present experiments due to their high clay and silt content compared to those cited; short duration of the experiments with huge unsaturation further helped to capture a higher amount of organic carbon curbing its losses through oxidation. Our findings are in agreement with Hassink and Whitmore (1997) who reported that a greater conversion efficiency of the added carbon (△SOC/△C input) to SOC, when carbon content lies far from the carbon saturation level. Among the sites, Gazipur and Aduthurai had higher total clay+silt content (69.3 and 82 %) and thus a higher carbon saturation deficit which may subsequently trigger greater carbon stabilization; while, Karnal site had a lower clay+silt, and thus expected a lower carbon stabilization rate (Stewart et al., 2009). However, the low conversion rate, in spite of having higher clay+silt in Patna was difficult to explain (Supplementary table 1).

Among the scenarios, stabilization of applied carbon input was found in the order of S4 (11.7 %) > S3 (10.6 %) > S2 (7.8 %) irrespective of locations (Table 5). This happened because of the fact that S4 had a maize-based cropping system. The added maize residue was more resistant (Abiven et al., 2011) to microbial decomposition and thus would facilitate a sharp build-up of TOC/WBC (Buranov and Mazza, 2008). Again, inclusion of legumes in summer season, especially mungbean and cowpea, and incorporating their residues was further expected to support a higher TOC/WBC in S4, S3 and S2 over S1 (Hazra et al., 2020); because legumes with its fixed nitrogen (Ladha et al., 2022) and residues with a relatively narrow C:N ratio decomposed quickly in soil enhancing soil microbial activity and facilitating conversion of crop residue carbon into SOC. Additionally, CT practice under S1 exposed soil aggregates to air, favouring higher carbon oxidation from soil; in contrast, ZT under S3 and S4 reduced the exposure of soil to air which ultimately helped to cut down the CO_2 emission and helped in higher TOC/WBC build-up (Corsi et al., 2012). All these favourable conditions have increased the conversion and subsequently the TOC/WBC significantly. Critical analysis of the performance of CA-based system in accruing carbon in soils across the world showed significant increase in its content in soils (Lessmann et al., 2022). Further rainfall and irrigation water management also play an important role in this conversion. Among the platforms, Aduthurai, Gazipur and Patna received a good amount of annual rainfall; whereas in Karnal site maintenance of good irrigation water management practices created a congenial soil moisture regime for carbon retention and accumulation (Matus, 2021).

Table 5
Stabilization rate of additional carbon input to TOC over S1 at different scenarios of all the sites.

Scenarios and locations		Δ Carbon input (over S1)	Δ TOC (over S1)	% C-Stabilization	Location average	Scenarios average
Karnal	S2	9.18	0.69	7.52	**7.7**	S2–7.8
	S3	10.29	0.83	8.06		S3–10.6
	S4	13.45	1.02	7.58		S4–11.7
Patna	S2	4.74	0.24	5.06	5.8	
	S3	7.12	0.54	7.58		
	S4	21.60	1.00	4.63		
Aduthurai	S2	4.34	0.41	9.45	12.1	
	S3	4.60	0.48	10.43		
	S4	4.34	0.71	16.36		
Gazipur	S2	3.08	0.28	9.09	14.7	
	S3	3.70	0.62	16.71		
	S4	5.20	0.95	18.46		

Where, S1: Conventional rice-wheat or rice-rice system with conventional farmer's practice; S2: Partial CA based rice-wheat-mungbean or rice-rice-blackgram or rice-rice-fallow system with best management practice (BMPs); S3: Full CA based rice-wheat-mungbean or rice-rice-blackgram or rice-rice-fallow with BMPs; S4 – Full CA based maize-wheat-mungbean or rice-potato – maize + mungbean or maize-rice-maize or rice-potato- maize system with BMPs.

4.3. Does CA facilitate mobilization of added C into pools of longer residence time?

The allocation of stabilized carbon was found more in passive pool and less in active pool (Fig. 1). This allocation of carbon was related to its biochemistry of substrate. Among all the locations, rice, wheat and maize crops were cultivated and their biomass incorporated; all the crop residues contain relatively higher proportions of lignin, cellulose and hemicellulose content. Our results showed that nearly 60 % of build-up carbon was stabilized in passive pool specially under S4 and S3 irrespective of locations. These conversion of C in soils thus indicate a win-win situation for S4 and S3 scenarios; they (S4 and S3) had not only a higher amount of SOC and facilitated a higher conversion of crop residue carbon to SOC, but also had a higher amount of it (C) found way to pools of longer residence time for its eventual sequestration in soils. Besides, we computed recalcitrance index (RI), and lability index of SOC and also carbon management index (CMI) for the scenarios. Corroborating our results, mentioned above, a higher recalcitrant index (RI), particularly at surface layer, was found in soils under CA-based scenarios (S4 > S3 > S2) over the conventional one (S1), with no definite trend for lability index (LI) (Supplementary Table 3). Further, a higher CMI in CA-based scenarios (S4 > S3 > S2) indicated that CA coupled with best management practices (balanced fertilization) and diversified cropping system offer an edge over best management practices with partial CA (Supplementary Table 3), and its value more than 100 designates sustainable management practices indicating systems ability to maintain vis-à-vis restore carbon content and support crop productivity for long run. Earlier, scenarios S3 and S4 were found to have higher yield, lower GWPi and also lower energy requirements (Ladha et al., 2016). A higher SOC and that too in pools of longer residence time under S4 and S3 make

those two systems fundamentally robust and sustainable than the S1 compared in the study.

4.4. Optimizing the systems performance and identification of best management practices

It is always better to evaluate a system in quantitative approach rather qualitative terms. Linear scoring technique was widely used in soil quality estimation (Andrews et al., 2002), after the formation of minimum dataset scoring was done and their summed-up values ultimately govern overall soil quality. Liebig et al. (2001) employed linear scoring techniques to evaluate relative sustainability of agricultural management systems within the various corn based agro-ecosystem. Similar to that, our important variables were already identified, hence scoring was done to determine overall system score and better score were obtained under CA based S4 and S3 over other two scenarios. Our findings thus proved that CA with diversified cropping system performed the best (S4), closely followed by CA with dominant cropping system in rotation with legume (S3), and partial CA with dominant cropping system in rotation with legume (S2) compared to the existing farmers practice (Ladha et al., 2016). However, long-term experiments across the sites with the studied objectives would have given robustness to the recommendations.

5. Conclusions

The study revealed a positive response of total organic carbon (TOC) and Walkley-Black carbon (WBC) pools across various agroecological zones in South Asia to conservation agriculture (CA) practices, demonstrating a consistent increase compared to conventional agriculture methods. Both active and passive carbon pools exhibited increment under CA practices, indicating enhanced microbial activity, improved nutrient cycling, and increased soil carbon sequestration. Systems implementing full CA with best management practices (BMPs) (S3) and BMPs combined with crop diversification (S4) displayed favourable characteristics for carbon economy, including balanced carbon distribution (reflected in higher Carbon Management Index - CMI), increased carbon stocks, and greater carbon stabilization, particularly within passive pools. This trend contributes to the resilience and sustainability of agricultural systems, fostering higher crop productivity (reflected in higher Resource Use Efficiency - REY) while also delivering environmental co-benefits, such as reduced Global Warming Potential intensity (GWPi) and energy consumption. Consequently, these practices are recommended for wider adoption and promotion in the studied regions, offering a pathway toward enhanced agricultural sustainability.

Declaration of competing interest

Authors declare no potential conflict of interest.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.geodrs.2024.e00870.

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